Practical Deep Learning

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Table of Contents

Learning Outcomes

- Build and train a convolutional network in PyTorch
- Make a training run reproducible: seed every generator, control determinism, and checkpoint enough state to resume
- Implement efficient data handling techniques using PyTorch's Dataset and DataLoader classes
- Apply appropriate data augmentation strategies based on error analysis
- Utilize transfer learning effectively for different scenarios and dataset sizes
- Understand ensemble methods to improve model performance
- Apply regularization techniques like Dropout and Batch Normalization
- Design a complete deep learning training workflow from initial setup to inference
- Choose and interpret evaluation metrics that suit the task and the class balance

- Practical Implementation of Deep Learning algorithms is just as much an art as it is a science.
- The main takeaway is not to start from scratch but rather to build on top of previous knowledge.
- Today, we will look at some important tools used in the practical implementation of Deep Learning algorithms.

CNN: **Implementing CNNs in PyTorch**

PyTorch Implementation of CNNs

```
import torch
import torch.nn as nn
import torch.nn.functional as F
class SimpleCNN(nn.Module):
   def __init__(self):
     super(SimpleCNN, self).__init__()
     self.conv1 = nn.Conv2d(3, 32, kernel_size=3, padding=1)
     self.pool = nn.MaxPool2d(2, 2)
     self.conv2 = nn.Conv2d(32, 64, kernel_size=3, padding=1)
     self.fc1 = nn.Linear(64 * 8 * 8, 128) # assumes 32x32 input (e.g.  CIFAR-10)
     self.fc2 = nn.Linear(128, 10) # assuming 10 classes
   def forward(self, x):
     x = self.pool(F.relu(self.conv1(x)))
     x = self.pool(F.relu(self.conv2(x)))
     x = x.view(-1, 64 * 8 * 8)
     x = F.relu(self.fc1(x))
     x = self.fc2(x)
     return x
```

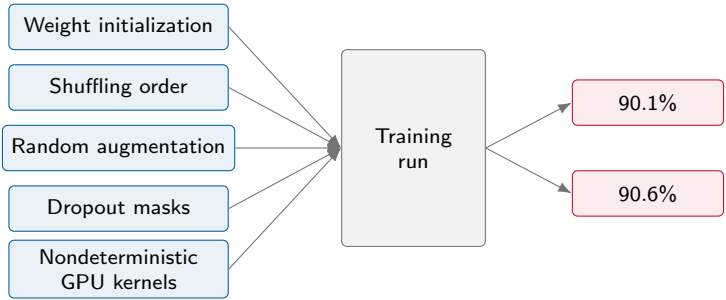
PyTorch Implementation of CNNs (cont.)

```
import torch.optim as optim
model = SimpleCNN()
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=0.001)
# Training loop
for epoch in range(10):
   for images, labels in dataloader:
     optimizer.zero_grad()
     outputs = model(images)
     loss = criterion(outputs, labels)
     loss.backward()
     optimizer.step()
```

Reproducibility

- You add an augmentation, retrain, and validation accuracy moves from 90.1% to 90.6%. Did the augmentation help?
- You cannot answer that until you know how much that number moves when you change **nothing at all**.
- Three things break when runs are not reproducible:
  - You cannot tell a real improvement from noise.
  - A colleague runs your code and gets a different number.
  - You come back in six months and cannot recover your own result.
- The fix is cheap: a handful of lines at the top of the script, plus the discipline to record what you ran.

Where the Randomness Comes From



- The first four all draw from a random number generator, so seeding fixes them.
- The last one does not: some GPU kernels sum partial results in whatever order the threads finish, and floating-point addition is not associative. It needs a separate switch.

Seeding: Three Generators, Not One

```
import random
import numpy as np
import torch
def set_seed(seed=42):
   random.seed(seed) # Python's own RNG
   np.random.seed(seed) # NumPy's RNG
   torch.manual_seed(seed) # CPU and every GPU
set_seed(42) # call this before you build the model
```

- Three libraries, three independent generators. Your augmentation pipeline may use any of them, so seed all three.
- `torch.manual_seed` ‘‘sets the seed for generating random numbers on all devices’’, so a separate `torch.cuda.manual_seed_all` is redundant.
- Seed before constructing the model: weight initialization is the first thing that consumes random numbers.

[0]Quotation and API from the PyTorch Reproducibility notes and `torch.manual_seed` documentation

The One Everybody Forgets: DataLoader Workers

- With `num_workers > 0` the DataLoader forks background processes, and **each worker gets its own copy of the RNG state**.
- Seeding the main process does not reach them, so your random crops and flips are unseeded even though everything else is.
- Two arguments fix it: `worker_init_fn` reseeds each worker, and `generator` pins the shuffling order in the main process.

The One Everybody Forgets: DataLoader Workers (cont.)

```
def seed_worker(worker_id):
    worker_seed = torch.initial_seed() % 2**32
    np.random.seed(worker_seed)
    random.seed(worker_seed)
g = torch.Generator()
g.manual_seed(42)
train_loader = DataLoader(train_set, batch_size=64,
                shuffle=True, num_workers=4,
                worker_init_fn=seed_worker, generator=g)
```

- PyTorch already gives each worker a distinct torch seed derived from g; seed_worker passes that same value on to NumPy and to Python's random.
- This is the single most common reason a ''seeded'' training script still wanders.

[0]Code follows the PyTorch Reproducibility notes, *DataLoader* section

Determinism Switches

```
import os
os.environ["CUBLAS_WORKSPACE_CONFIG"] = ":4096:8"
import torch # set the variable before torch loads cuBLAS
torch.use_deterministic_algorithms(True)
torch.backends.cudnn.deterministic = True
torch.backends.cudnn.benchmark = False
```

- `use_deterministic_algorithms(True)` makes PyTorch pick a deterministic kernel wherever one exists.
- `cudnn.benchmark = False` switches off the autotuner, which otherwise times several convolution algorithms and may choose a different one from run to run.
- Several cuBLAS reductions are nondeterministic from CUDA 10.2 onwards unless `CUBLAS_WORKSPACE_CONFIG` is set to `:4096:8` or `:16:8`.

Determinism Is Not Free

- Some operations have **no** deterministic implementation on CUDA: `AvgPool3d`, `AdaptiveAvgPool2d/3d`, `MaxUnpool`, reflection padding, `NLLLoss` and `CTCLoss` are among them.
- Under `use_deterministic_algorithms(True)` they raise a `RuntimeError` instead of quietly staying random. That is a feature: it names the layer that is the source of the wobble.
- Deterministic kernels are usually slower, and losing the cuDNN autotuner costs more still.

Situation	**Turn determinism on?**
Chasing a bug you cannot reproduce	Yes, fully
Numbers going into a report	Yes, plus a seed sweep
Comparing two ideas	Fix the seed in both arms
A long production run	No, keep `benchmark=True`

Same Seed Does Not Mean Same Number

- The PyTorch documentation is blunt about this: "Completely reproducible results are not guaranteed across PyTorch releases, individual commits, or different platforms. Furthermore, results may not be reproducible between CPU and GPU executions, even when using identical seeds."
- So a seeded run is reproducible on *that* machine, with *that* software stack. Change any of the following and the digits move:
  - a different GPU architecture, driver, or cuDNN build;
  - a different PyTorch or CUDA version;
  - a different `num_workers` — the batches are the same, but a different worker draws the augmentation for each sample, so the images are not.
- What you should aim for is **stochastic equivalence**: a run you can repeat exactly on your own machine, and a *distribution* of results you can quote elsewhere — not bit-equality.

[0]Quotation: PyTorch Reproducibility notes

How Much Can the Seed Alone Move a Score?

Picard trained the same CIFAR-10 network thousands of times, changing only `torch.manual_seed`.

CIFAR-10, ResNet9	Seeds	Mean $\pm$ std	Min	Max
Long training (converged)	500	90.70 ± 0.20	90.14	91.41
Short training	10^4	90.02 ± 0.23	89.01	90.83

- Across 10^4 seeds the luckiest and unluckiest runs are 1.82 points apart — larger than plenty of published "improvements".
- On ImageNet with a pretrained ResNet50 (50 seeds) the effect shrinks but does not vanish: mean around 75.5%, standard deviation about 0.1%, a min-to-max gap of roughly 0.5%.
- **Practical rule:** if your improvement is smaller than the spread you get from reseeding, you have not measured an improvement yet. Run three to five seeds and compare the means.

[0]Numbers from Picard, "torch.manual_seed(3407) is all you need", arXiv:2109.08203v2, 2023, Tables 1–2

Pin the Environment, Not Just the Seed

- A seed is worthless if the library that consumes it changed underneath you. Record the stack next to the result:
  - the exact package versions (`pip freeze > requirements.txt`, a `uv.lock`, or `conda env export`);
  - the git commit of the training code;
  - the GPU model, driver, CUDA and cuDNN versions.

```
print(torch.__version__, torch.version.cuda)
print(torch.backends.cudnn.version())
print(torch.cuda.get_device_name(0))
```

- Three lines at the start of every run, written into the log. It costs nothing and it is the first thing you will want when a number stops replicating.
- Containers, lockfile-driven CI, experiment trackers and model registries take this much further --- that is the subject of the MLOps deck in the Introduction to AI course, not this one.

Checkpoint Discipline

- A file containing only `model.state_dict()` is enough for inference and useless for resuming: the optimizer's momentum buffers and the scheduler's step count are gone.
- Save the whole training state, and the configuration that produced it, in one object.

Checkpoint Discipline (cont.)

```
torch.save({
   "epoch":  epoch,
   "model_state_dict":  model.state_dict(),
   "optim_state_dict":  optimizer.state_dict(),
   "sched_state_dict":  scheduler.state_dict(),
   "best_val_acc":  best_val_acc,
   "seed":  seed,
   "config":  config, # lr, batch size, augment
   "torch_version":  torch.__version__,
}, "resnet9_seed42_ep12_acc9071.pt")
```

- The file name carries the metadata you will search on later: architecture, seed, epoch, score. model_final_v2_REAL.pt does not.
- Keep the payload to tensors and plain Python types --- torch.load now defaults to weights_only=True, which refuses arbitrary pickled objects.

[0]Checkpoint keys follow the PyTorch Saving and Loading Models tutorial

Resuming a Run Correctly

```
ckpt = torch.load(path, weights_only=True)
model.load_state_dict(ckpt["model_state_dict"])
optimizer.load_state_dict(ckpt["optim_state_dict"])
scheduler.load_state_dict(ckpt["sched_state_dict"])
start_epoch = ckpt["epoch"] + 1
model.train() # dropout and BN back into train mode
```

- Forgetting `model.train()` after loading leaves dropout off and batch normalization on stored statistics --- training continues, badly, and nothing warns you.
- Rebuild the model and optimizer objects first; `load_state_dict` fills an existing object, it does not construct one.

Resuming a Run Correctly (cont.)

- Resuming does not give you the run you would have had without interrupting it. Reseeding at startup rewinds the generators, so the data order for epoch 12 is the order you saw in epoch 0.
- If the continued run has to match, save the generator states too:

```
# add to the checkpoint dictionary
"torch_rng":  torch.get_rng_state(),
"cuda_rng":  torch.cuda.get_rng_state_all(),
"numpy_rng":  np.random.get_state(),
"python_rng":  random.getstate(),
```

- Restore them with the matching setters (torch.set_rng_state, np.random.set_state, random.setstate) instead of calling set_seed again.
- For most coursework this is overkill --- but know that it is the reason a resumed run and an uninterrupted run rarely land on the same number.

Reproducibility Checklist

- ✓ Seed `random`, `numpy` and `torch` before building the model.
- ✓ Pass `worker_init_fn` *and* `generator` to every DataLoader.
- ✓ Turn on `use_deterministic_algorithms` and the cuDNN flags while debugging; turn them off for long runs.
- ✓ Log the package versions, git commit and GPU alongside every result.
- ✓ Checkpoint model, optimizer, scheduler, epoch, seed and config together, with a name you can read.
- ✓ Report more than one seed before you claim an improvement.
- ▶ The NeurIPS reproducibility checklist generalizes this to a whole paper: code, data splits, hyperparameter search ranges, number of runs, and error bars.

[0]Pineau et al., "Improving Reproducibility in Machine Learning Research", JMLR 22(164):1–20, 2021

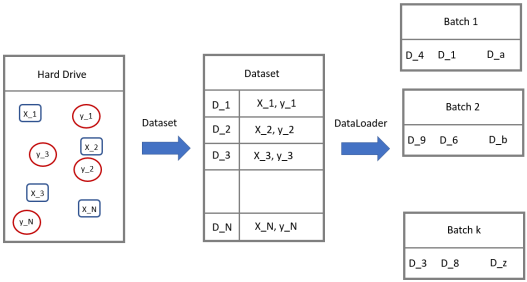
DataLoaders

- As we have previously established, Deep Learning has been made possible by large amounts of data and computational resources.
- An important aspect to keep in mind is the data handling:
  - How do we handle large amounts of data?
  - How do we read different components of data (from possibly different parts of our hard drive) and provide it to our training algorithms?
  - How do we feed this data to SGD algorithms in a streamlined manner?
- PyTorch provides Dataset and DataLoaders to handle data in an efficient manner.
- We will subclass `Dataset` to describe our own data, then hand it to a `DataLoader` for batching and shuffling.

DataLoaders (cont.)

- Datasets in PyTorch: The `torch.utils.data.Dataset` class is an abstract base class used to define and manage datasets for training and evaluation.
- Custom Dataset Creation: You can create a custom dataset by subclassing `Dataset` and implementing
  - `__len__()` (returns dataset size) and
  - `__getitem__()` (fetches a single data sample).
- DataLoaders for Batching: `torch.utils.data.DataLoader` is used to load data in mini-batches, shuffle data, and utilize multiprocessing for efficiency.
- Built-in Datasets: PyTorch provides datasets in `torchvision.datasets`, making it easy to load common datasets like MNIST and CIFAR-10.

DataLoaders (cont.)



A Custom Dataset and DataLoader

```
from torch.utils.data import Dataset, DataLoader
from torchvision import transforms
from PIL import Image
class ImageDataset(Dataset):
   def __init__(self, paths, labels, transform=None):
     self.paths = paths
     self.labels = labels
     self.transform = transform
   def __len__(self):  # how many samples the dataset holds
     return len(self.paths)
   def __getitem__(self, idx):  # fetch one sample
     image = Image.open(self.paths[idx]).convert("RGB")
     if self.transform:
       image = self.transform(image)
     return image, self.labels[idx]
```

A Custom Dataset and DataLoader (cont.)

```
to_tensor = transforms.Compose([transforms.ToTensor()])
train_set = ImageDataset(paths, labels, transform=to_tensor)
train_loader = DataLoader(train_set, batch_size=64, shuffle=True,
                          num_workers=4)
```

- Only `__len__()` and `__getitem__()` are ours to write; `__getitem__()` reads one sample from disk, so nothing beyond the current batch is held in memory.
- The DataLoader does the rest: it groups samples into batches, reshuffles them each epoch, and reads ahead in num_workers background processes so the GPU is not left waiting.

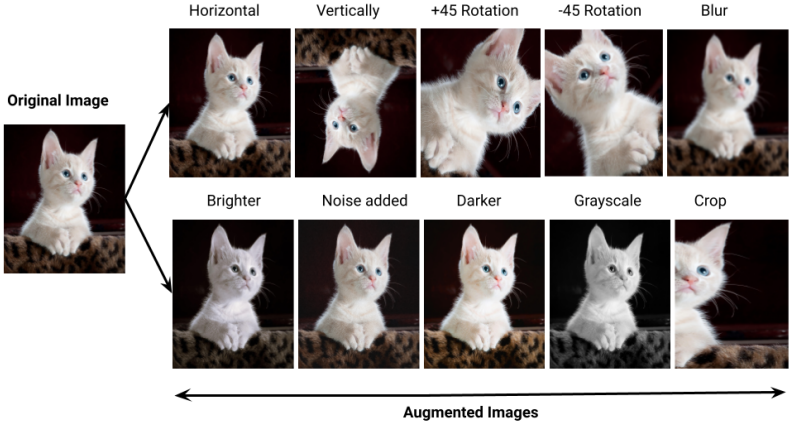
Data Augmentation

- Data is the fundamental building block of any machine learning algorithm
- In several applications we don't have access to unlimited data
- So we use Data Augmentation techniques to improve the performance of our models
- Note: It is better to spend time on data rather than fine-scale architecture search in deep learning

Data Augmentation (cont.)

- Create virtual training samples
  - Horizontal flip
  - Random crop
  - Color casting
  - Geometric distortion
  - Translation
  - Rotation

Data Augmentation



[0]Figure: Pranjal Ostwal, "Data Augmentation for Computer Vision" (Medium)

Data Augmentation

[0]Figure: albumentations transform demo (https://github.com/albumentations-team/albumentations; now archived, successor: AlbumentationsX). Library: Buslaev et al., "Albumentations: Fast and Flexible Image Augmentations", *Information* 11(2):125, 2020.

Data Augmentation (cont.)

- But, there are many types of augmentations. How do we choose the right ones for our task?

Data Augmentation (cont.)

- But, there are many types of augmentations. How do we choose the right ones for our task?
- **Answer: Error Analysis** – Identify model weaknesses and apply augmentations that address those issues.

Error Analysis for Data Augmentation

- **Steps:**
  1. Train a baseline model.
  2. Make predictions on validation data.
  3. Inspect the worst predictions to identify model weaknesses.
  4. Apply relevant augmentations to address these issues.
- **Examples:**
  - **Failure with small objects** → Use *Scale Augmentation*.
  - **Failure with different colors/environments** → Use *Color Augmentations*.
  - **Failure with rotated images** → Use *Rotation Augmentations*.
  - **Failure with blurry images** → Use *Blur Augmentations* (e.g., GaussianBlur, MotionBlur).
  - . . . etc.

Data Augmentation

- **Note**: Data augmentation is applied only to the training data.
- Applying it to validation/test data **directly** would lead to **incorrect evaluation and misleading performance metrics**.

- However, there is a special inference technique called **Test-Time Augmentation (TTA)** where multiple augmented versions of the same image are passed through the model separately, and the predictions are averaged to improve accuracy.

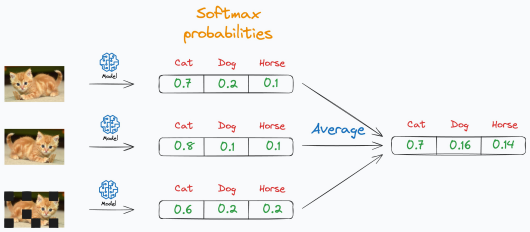


Figure 1: Test-time Augmentation

[0]Figure: Avi Chawla & Banias Baabe, "Train and Test-time Data Augmentation", Daily Dose of Data Science, 2024.

Data Augmentation

- TTA is an advanced technique.
- Applying it requires **custom scripts** to generate augmented versions of test images, pass them through the model, and aggregate predictions.

Transfer Learning

- **What is Fine-Tuning?**
  - A strategy in **transfer learning** where a pre-trained model is adapted to a new task.
  - Instead of training from scratch, we start from a model already trained on a related task.
- **Why use Fine-Tuning?**
  - Saves time and computational resources.
  - Improves performance, especially with limited data.

When to fine-tune your model?

- New dataset is small + similar distribution to original dataset:
  - Freeze (or partially freeze) feature extraction layers and fine-tune the classifier.
- New dataset is small + different distribution to original dataset:
  - Use the pretrained network as a generic feature extractor and train a light classifier on top (e.g., SVM).
  - In modern practice, you might also freeze earlier layers and selectively fine-tune later layers.
- New dataset is large, regardless of the original data distribution:
  - Fine-tune the entire network (both feature extractor and classifier).

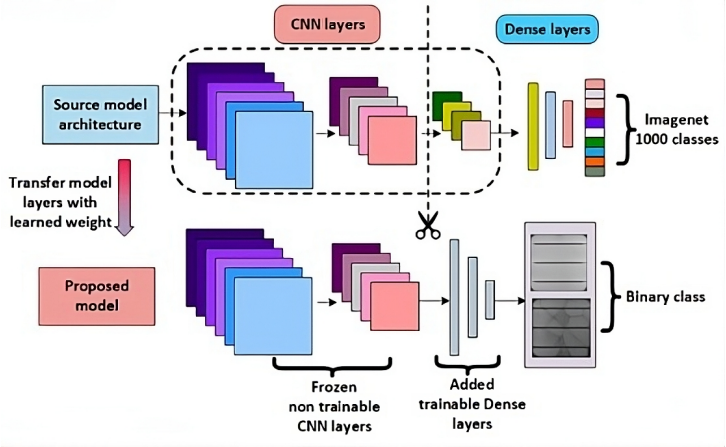


Figure 2: Fine-tuning a pretrained ImageNet model for a new task

[0]Figure: Rahman, Tabassum, Haque, Nishat, Faisal & Hossain, "CNN-based Deep Learning Approach for Micro-crack Detection of Solar Panels", STI 2021, Fig. 3.

- No single model is perfect. Different models make different types of errors.
- Let's visualize the predictions of two different models:

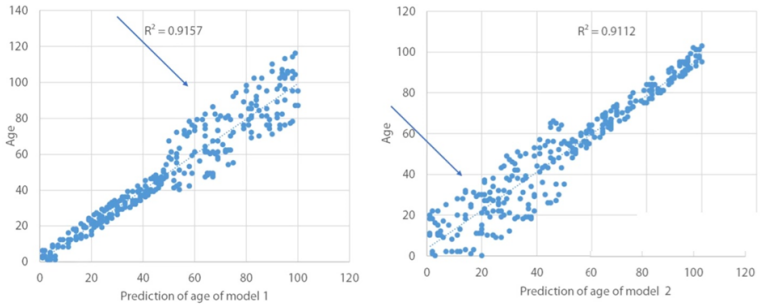


Figure 3: Predictions of Model 1 and Model 2

[0]Figure: Coursera / HSE, "How to Win a Data Science Competition: Learn from Top Kagglers", week 4 ensembling lecture.

- Combining predictions of diverse models can reduce errors and improve accuracy. This technique is called **Ensembling**.

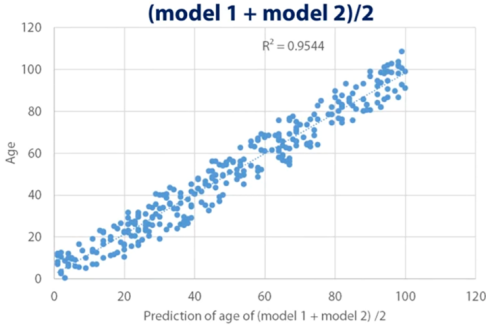


Figure 4: Averaged Predictions: Better Correlation

[0]Figure: Coursera / HSE, "How to Win a Data Science Competition: Learn from Top Kagglers", week 4 ensembling lecture.

- Diverse models are key to effective ensembling. Here are two strategies:
  - **Bagging (Bootstrap Aggregating):** Train multiple models on different subsets of data (e.g., Random Forest model).

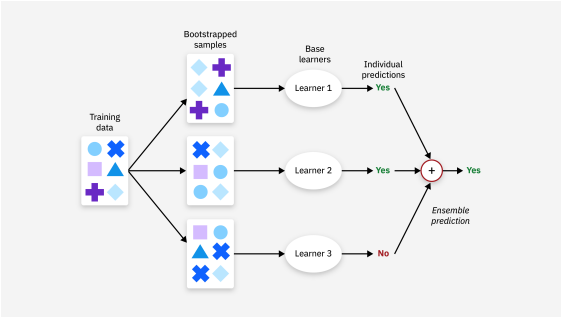


Figure 5: Bagging example

[0]Figure: IBM, https://www.ibm.com/think/topics/bagging

- Boosting:
  - Train models sequentially, where each model focuses on the errors of the previous one (e.g., AdaBoost, Gradient Boosting).

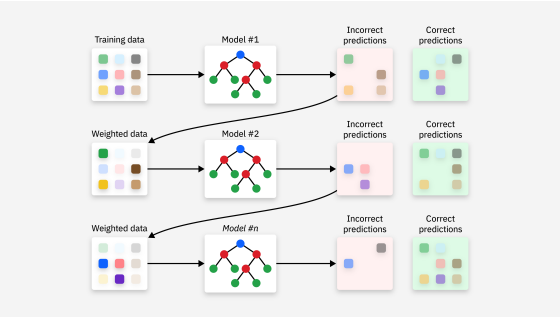


Figure 6: Boosting Example

[0]Figure: IBM, https://www.ibm.com/think/topics/boosting

Ensembling

- These strategies introduce diversity in models, but how can we combine their predictions?

Ensembling

- We can combine classifiers' predictions in two ways:
  - **Hard Voting:**
    - Each model votes for a class.
    - The class with the majority of votes is selected.
  - **Soft Voting:**
    - Average the predicted probabilities from each model.
    - The class with the highest average probability is selected.
- For regressors: take the average of predictions from all models.

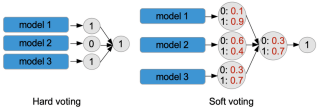


Figure 7: Illustration of Hard and Soft Voting for Classifiers

[0]Figure: Bulaty Tauil et al., Rev. Observatorio de la Economía Latinoamericana 22(5), 2024, Fig. 2.

Ensembling

- Teamwork is the best policy.
- We can train multiple networks for the same task then ensemble to get better results.

Ensembling - A simple Analysis

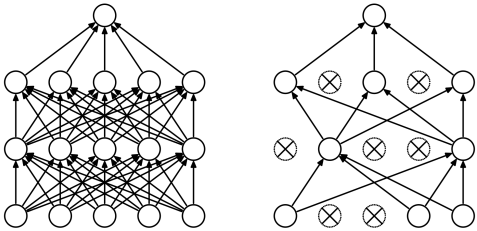
- Let's assume that we have a test dataset with N elements and an ensemble of M models.
- Also assume that each model in the ensemble errs independently on an image with the same probability, denoted by e.
- For example, assume $M = 3$ and $e = 0.01$.
- Then the probability of error of the label for the max-voting ensemble will be

$$p_{\text{ens}} = 1 - (1 - e)^3 - \binom{3}{2}(1 - e)^2 e$$

- For the above example $p_{\text{ens}} = 0.0003$, which is significantly lower than a single model.

Dropout

- **Dropout** is a regularization technique used to prevent overfitting in neural networks.
- During training, it randomly drops (sets to 0) a fraction of neurons in each layer based on a specified probability for every forward pass.



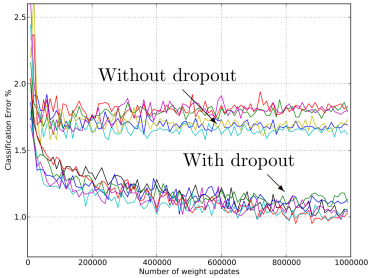
(a) Standard Neural Net (b) After applying dropout.

[0]Srivastava JMLR 2014

Dropout

- Why is dropping neurons randomly useful?
  1. By dropping a subset of neurons during each forward pass, the network learns **not to rely too heavily on specific connections**, encouraging the network to use more connections.
  2. Dropout acts as an **efficient ensemble** of multiple smaller networks, each trained on a random subset of neurons.
  3. More Connections + Diverse Ensemble = **less overfitting**.

Dropout

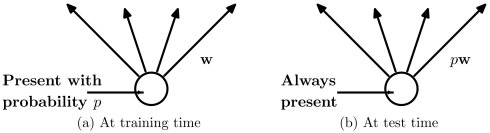


Model	Top-1 (val)	Top-5 (val)	Top-5 (test)
SVM on Fisher Vectors of Dense SIFT and Color Statistics	-	-	27.3
Avg of classifiers over FVs of SIFT, LBP, GIST and CSIFT	-	-	26.2
Conv Net + dropout (Krizhevsky et al., 2012)	40.7	18.2	-
Avg of 5 Conv Nets + dropout (Krizhevsky et al., 2012)	38.1	16.4	16.4

Table 6: Results on the ILSVRC-2012 validation/test set.

[0] Srivastava JMLR 2014

Dropout



(a) At training time (b) At test time

- We drop neurons during training, but what should we do during inference?
- **During inference:** We use all neurons.

But...

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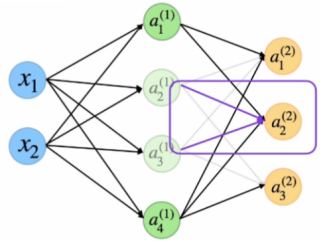
Dropout



Assume we have present probability $p = 0.5$

During testing, when nodes are present again, activations are $2 \times$ as large!

[0] Deep Learning Fundamentals - Lightning AI

Dropout

- **Problem:** Having inconsistent values during inference compared to training can cause the network to produce worse results.
- **Solution:** Scale the output during inference by the **keep probability** p (the chance a neuron stays active) to keep the same expected activation as in training.
- **Example:**
  - **During training (keep probability $p = 0.1$)**:
    - Activation value $= 2$
    - Active only 10% of the time
    - $\Rightarrow$ Effective contribution $= 2 \times 0.1 = 0.2$
  - **During inference**:
    - Always active ($\Rightarrow$ Activation value $= 2$)
    - Scale by $p = 0.1 \Rightarrow 2 \times 0.1 = 0.2$
  - Result: Consistent activation scale between training and inference.

Dropout

- In PyTorch, dropout behavior is controlled by the model's mode:

```
# Training Mode:  Enables dropout
model.train()
output = model(input)

# Evaluation Mode:  Disables dropout (no scaling needed;
# PyTorch already scales by 1/(1-p) during training)
model.eval()
output = model(input)
```

- **Note:** `nn.Dropout(p)` takes the *drop* probability, so a keep probability of 0.1 is written `nn.Dropout(0.9)`.

Dropout in Practice

- **How much to drop:** a keep probability of $p = 0.5$ on the hidden layers is close to optimal across most networks and tasks, and the test error stays flat for $0.4 \leq p \leq 0.8$. Input layers want a higher keep probability, around 0.8.
- **It costs training time:** a dropout network typically takes 2 to 3 times longer to train than the same architecture without it, because each update is computed on a different thinned network and so is much noisier.
- **It does not always help:** with very little training data the network can memorise the training set even through the dropout noise. On MNIST subsets of 100 and 500 images, dropout gave no improvement.

[0] Srivastava JMLR 2014, Sections 7.3, 7.4 and 11

Batch Normalization

- **Key Insight**: Networks train better when inputs are normalized
- So why not normalize intermediate layers too?
- **Problem**: forcing every layer to output zero-mean, unit-variance values might be too restrictive (not always optimal).
- **Solution**: Let the network learn if it wants normalization!

Batch Normalization

For each feature in a layer:

1. First normalize: $\hat{x} = \frac{x-\mu}{\sqrt{\sigma^2+\epsilon}}$
   - μ: mean across batch dimension
   - σ^2: variance across batch dimension
   - Computed separately for each feature/channel
2. Then give the network control: $y = \gamma\hat{x} + \beta$
   - γ (scale): Can amplify or reduce the normalized values
   - β (shift): Can move the values away from zero
   - These are learned during training like normal weights!

Why γ and β Matter

- After normalization, outputs are always zero-mean, unit-variance
- But this might not be optimal for every layer!
- γ and β let each layer learn:
  - γ: "How much variance do I want?"
  - β: "What should my mean activation be?"
- Two extremes the network can learn:
  - "Keep normalization": $\gamma \approx 1$, $\beta \approx 0$
  - "Undo normalization": γ and β restore original scale and shift
- Network learns what's best for each layer!

Mathematically, for batch size N, at each feature/channel j:

$$\mu_j = \frac{1}{N}\sum_{i=1}^{N} x_{i,j} \quad \text{(batch mean)}$$

$$\sigma_j^2 = \frac{1}{N}\sum_{i=1}^{N}(x_{i,j} - \mu_j)^2 \quad \text{(batch variance)}$$

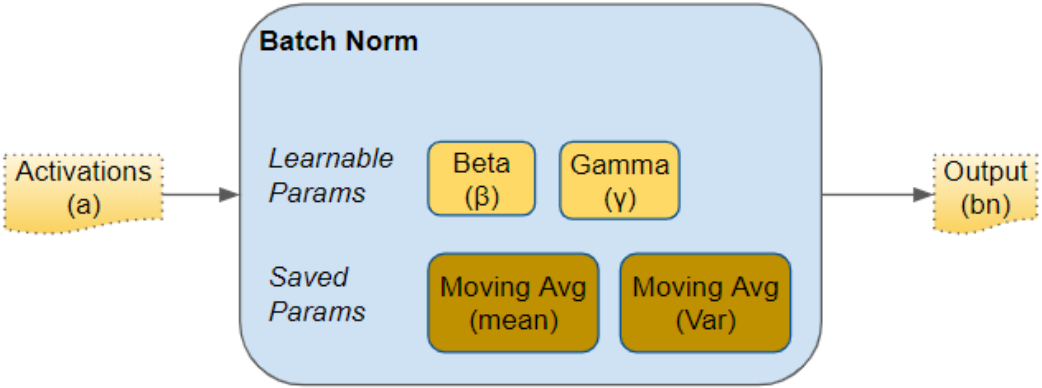
$$\hat{x}_{i,j} = \frac{x_{i,j} - \mu_j}{\sqrt{\sigma_j^2 + \epsilon}} \quad \text{(normalize)}$$

$$y_{i,j} = \gamma_j \hat{x}_{i,j} + \beta_j \quad \text{(learnable transform)}$$

Batch Normalization

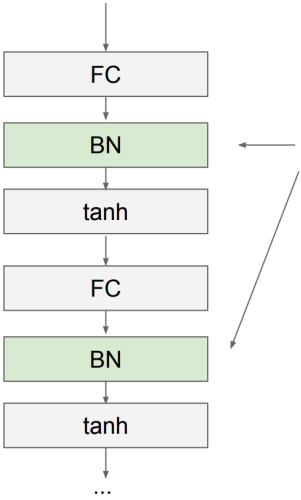
- During Training:
  - Use statistics from current batch
  - Keep running average of mean and variance (to use later in inference):
    $\mu_{running} = 0.9 \times \mu_{running} + 0.1 \times \mu_{batch}$
- During Inference (We can't depend on mini-batches):
  - Use stored running averages ($\mu_{running}$, $\sigma^2_{running}$)
  - These are fixed values from training
  - No batch statistics needed!

Batch Normalization



[0]Figure: Ketan Doshi, "Batch Norm Explained Visually" (towardsdatascience.com)

Batch Normalization



Usually inserted after Fully Connected or Convolutional layers, and before nonlinearity.

[0]Figure: CS231n, Stanford University

Batch Normalization

- In PyTorch, batch normalization behavior is controlled by the model's mode:

```
# Training Mode:  Use batch statistics
model.train()
output = model(input)

# Evaluation Mode:  Use running statistics
model.eval()
output = model(input)
```

Batch Normalization

- Advantages:
  - Makes deep networks much easier to train!
  - Improves gradient flow
  - Allows higher learning rates, faster convergence
  - Networks become more robust to initialization
  - Acts as regularization during training
  - Zero overhead at test-time: can be fused with conv!

[0]Slide based on Stanford CS231n by Fei-Fei Li, Yunzhu Li & Ruohan Gao

Batch Normalization

- Advantages:
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  - Improves gradient flow
  - Allows higher learning rates, faster convergence
  - Networks become more robust to initialization
  - Acts as regularization during training
  - Zero overhead at test-time: can be fused with conv!
- Disadvantages:
  - Behaves differently during training and testing: this is a very common source of bugs!

[0]Slide based on Stanford CS231n by Fei-Fei Li, Yunzhu Li & Ruohan Gao

Batch Normalization

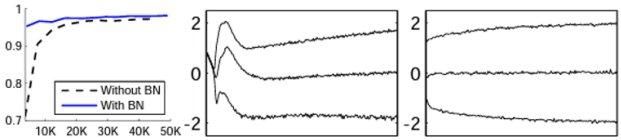

Figure 1: (a) *The test accuracy of the MNIST network trained with and without Batch Normalization, vs. the number of training steps. Batch Normalization helps the network train faster and achieve higher accuracy.* (b, c) *The evolution of input distributions to a typical sigmoid, over the course of training, shown as* $\{15, 50, 85\}$ *th percentiles. Batch Normalization makes the distribution more stable and reduces the internal covariate shift.*

[0] Ioffe and Szegedy 2015

Dropout and Batch Normalization Together

- Both reduce overfitting, so it is tempting to switch on both. They interfere.
- Dropout changes the variance of a unit between training and test time. Batch normalization stores the variance it measured during training and reuses that stored value at test time, so the two disagree about the same quantity.
- The disagreement compounds from layer to layer and shows up as a drop in test accuracy. It has been measured across a range of modern convolutional networks.
- **Practical rule:** put dropout after all batch normalization layers, usually in the fully connected head, rather than between a convolution and its normalization. Many modern architectures get their best results from batch normalization with no dropout at all.

[0]Li, Chen, Hu & Yang, "Understanding the Disharmony between Dropout and Batch Normalization by Variance Shift", CVPR 2019

- **Step 1: Initial Setup**
  - Start with a pretrained model and just finetune when possible; it saves time and improves results.
  - Define an initial architecture without regularization (e.g., dropout) or augmentations.
  - Set up validation strategy and choose an appropriate evaluation loss and metric.
  - Train the model to get a **baseline score**.

Full Training Workflow

- **Step 2: Improvement Process**
  - Overfitting is common at the start; use regularization techniques like:
    - Dropout, batch normalization,...
  - Perform error analysis to identify weaknesses and choose appropriate augmentations or preprocessing.
  - Tune hyperparameters: layers, epochs, learning rate, batch size, etc.
  - Optionally, use ensembling to boost scores (requires more resources).
- **Key Tip:** Track scores and improvements at every step to measure progress effectively.

Measuring an Improvement Against the Baseline

- The baseline from Step 1 is the thing every later run is compared to. Freeze it and keep its checkpoint.
- Change **one** thing at a time, and hold the seed fixed across both arms, so the difference you measure is the change and not the draw.
- Before believing a gain, find out how big the noise is: rerun the baseline with a different seed and nothing else altered.
- If the improvement is smaller than that seed-only spread, it is not an improvement yet. Rerun both arms on three to five seeds and compare the means.

Keep a Run Log

- Every run should leave a row behind: what you changed, what it scored, and enough information to reproduce it.

Run	Commit	Seed	Change from baseline	Val F1
001	a3f19c	42	baseline, no augmentation	0.812
002	a3f19c	43	baseline, seed only	0.807
003	7d2e40	42	+ random crop and flip	0.841
004	91b6aa	42	+ dropout 0.3 in the head	0.838

- Run 002 is the one that makes the table worth keeping: reseeding alone moved the score by 0.005, so run 004 is noise while run 003 is real.
- A spreadsheet is enough for a course project, as long as you fill it in as you go rather than from memory afterwards.
- Tools such as TensorBoard, MLflow and Weights & Biases automate this and version the artifacts too — covered in the MLOps deck of the Introduction to AI course.

- **Step 3: Finalization**
  - Save the optimized model for deployment.
  - Use the model for inference in real-world applications.

Accuracy:

- The ratio of correctly predicted instances to the total instances.
- It is a common metric for classification tasks.
- Formula:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

- Where:
  - TP: True Positives
  - TN: True Negatives
  - FP: False Positives
  - FN: False Negatives
- Limitations:
  - It can be misleading in imbalanced datasets.
  - It does not provide information about the types of errors made.

CNN - Evaluation Metrics (cont.)

Precision:

- The ratio of correctly predicted positive instances to the total predicted positive instances.
- Formula:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

- It indicates how many of the predicted positive instances were actually positive.
- High precision means fewer false positives.
- Limitations:
  - It does not consider false negatives.
  - It can be misleading in imbalanced datasets.

Recall:

- The ratio of correctly predicted positive instances to the total actual positive instances.
- Formula:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

- It indicates how many of the actual positive instances were predicted correctly.
- High recall means fewer false negatives.
- Limitations:
  - It does not consider false positives.
  - It can be misleading in imbalanced datasets.

CNN - Evaluation Metrics (cont.)

F1 Score:

- The harmonic mean of precision and recall.
- Formula:

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

- It provides a balance between precision and recall.
- Useful for imbalanced datasets where one class is more important than the other.
- Limitations:
  - It can be misleading if the precision and recall are very different.
  - It does not provide information about the types of errors made.

CNN - Evaluation Metrics (cont.)

Confusion Matrix:

- A table that summarizes the performance of a classification model.
- It shows the true positive, true negative, false positive, and false negative counts.
- It provides a detailed breakdown of the model's performance.
- Useful for understanding the types of errors made by the model.
- Can be used to calculate other metrics like accuracy, precision, recall, and F1 score.
- Example:

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

- Where:
  - TP: True Positives
  - TN: True Negatives
  - FP: False Positives
  - FN: False Negatives

CNN - Evaluation Metrics (cont.)

mAP:

- mAP is calculated by averaging the average precision (AP) across all classes.
- AP is calculated by plotting the precision-recall curve and computing the area under the curve.
- mAP is particularly useful for evaluating models on datasets with multiple classes.
- It provides a comprehensive measure of the model's performance across all classes.
- Example:

$$\text{mAP} = \frac{1}{N} \sum_{i=1}^{N} \text{AP}_i$$

- Where:
  - N is the number of classes.
  - AP_i is the average precision for class i.
- mAP can be calculated at different IoU thresholds (Intersection over Union: the overlap between the predicted and true boxes; e.g., 0.5, 0.75) to evaluate the model's performance at different levels of overlap.

CNN - Evaluation Metrics (cont.)

Listing 1: Code snippet (PyTorch)

```
from torcheval.metrics.functional import multiclass_precision,
    multiclass_recall, multiclass_f1_score

p = multiclass_precision(preds, labels)
r = multiclass_recall(preds, labels)
f1= multiclass_f1_score(preds, labels)
```

References

These slides have been adapted from

- Fei-Fei Li, Yunzhu Li & Ruohan Gao, Stanford CS231n: Deep Learning for Computer Vision
- Sebastian Raschka, Lightning AI: Deep Learning Fundamentals
- PyTorch documentation: torch.utils.data
- PyTorch documentation: Reproducibility and Saving and Loading Models

Primary sources for the methods covered

- Srivastava, Hinton, Krizhevsky, Sutskever & Salakhutdinov, Dropout: A Simple Way to Prevent Neural Networks from Overfitting, JMLR 15, 2014
- Ioffe & Szegedy, Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift, ICML 2015
- Li, Chen, Hu & Yang, Understanding the Disharmony between Dropout and Batch Normalization by Variance Shift, CVPR 2019

References (cont.)

- Buslaev, Iglovikov, Khvedchenya, Parinov, Druzhinin & Kalinin, Albumentations: Fast and Flexible Image Augmentations, Information 11(2):125, 2020
- Pineau, Vincent-Lamarre, Sinha, Larivière, Beygelzimer, d'Alché-Buc, Fox & Larochelle, Improving Reproducibility in Machine Learning Research (A Report from the NeurIPS 2019 Reproducibility Program), JMLR 22(164):1–20, 2021 (arXiv:2003.12206v4)
- Picard, torch.manual_seed(3407) is all you need: On the influence of random seeds in deep learning architectures for computer vision, arXiv:2109.08203v2, 2023